COMPOUNDS OF INDUSTRIAL IMPORTANCE

PYQ • 2021 - 2026

1. 2026 **4 MARKS**

(A) Some hints about the compound X are given.

(i) Known as the king of chemicals.

(ii) It is prepared industrially through the contact process.

Write the answers to the questions given below.

(a) Write the ionisation equation of X.

(b) Write the name of the acid formed when X reacts with NaCl.

(c) What is the nature of the salt formed when X reacts with ammonium hydroxide ? Write the reason.

OR

(B) The chemical formula of calcium chloride is $CaCl_2$.

(a) Which alkali should be used to form this salt ?

(b) Write the ionisation equation of this alkali.

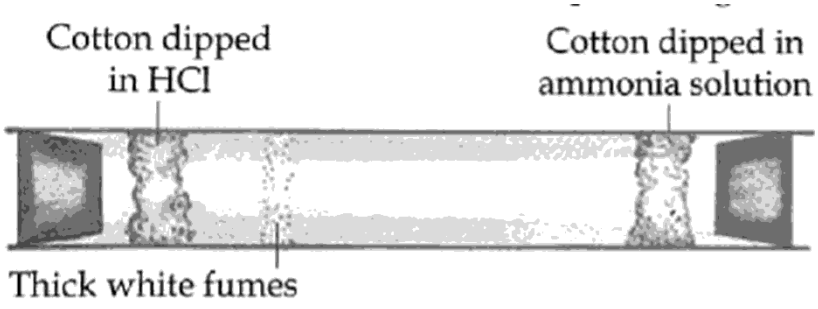
(c) Will calcium chloride undergo salt hydrolysis ? Why ?

2. 2026 **2 MARKS**

Observe the picture and answer the questions given below.

(a) Which substance is responsible for the thick white fumes ?

(b) Write the chemical equation of formation of this substance.



3. 2025 **4 MARKS**

A few drops of the solution 'X' is added to Magnesium sulphate solution taken in a test tube. A white precipitate is formed.

(a) What is 'X' ?

(b) What is the chemical name of white precipitate formed here ?

(c) What happens to the white precipitate when dilute hydrochloric acid is added ?

(d) Which type of salt is identified by this experiment ?

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4. 2025 **3 MARKS**

Two reversible reactions are given.

(I) $H_2(g) + I_2(g) \leftrightarrow 2HI(g) + Heat$

(II) $2SO_2(g) + O_2(g) \leftrightarrow 2SO_3(g) + Heat$

(a) Identify the reaction in which the pressure has no effect.

(b) How do the following factors influence the amount of product in the reaction (II) ?

(i) Increase the temperature (ii) Increase the pressure

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5. 2025 **1 MARK**

Which is the product obtained when SO_3 gas is dissolved in concentrated sulphuric acid ?

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6. 2024 **4 MARKS**

$2NO(g) + O_2(g) \leftrightarrow 2NO_2(g) + Heat$

(a) When does a reversible reaction attain equilibrium ?

(b) How do the following changes influence the amount of the product ?

(i) Increase in temperature

(ii) Increase in pressure

(iii) NO_2 is removed from the system.

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7. 2024 **3 MARKS**

Sulphuric acid is called the king of chemicals.

(a) Industrial preparation of sulphuric acid is known as _____ .

(b) Which is the catalyst used in this process ?

(c) When concentrated sulphuric acid is added to sugar, a black substance is formed. Explain the reason for this observation.

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8. 2024 **1 MARK**

When ammonium chloride is heated, a gas with basic nature is obtained. Write the name of the gas.

9. 2022 **4 MARKS**

(a) The industrial preparation of sulphuric acid is known as _____ .

(b) Which is the catalyst used in this process ?

(c) Take some sugar in a watch glass and add a few drops of concentrated sulphuric acid into it. What is your observation ? Which chemical property of sulphuric acid is shown here ?

10. 2022 **3 MARKS**

$N_2(g) + 3H_2(g) \leftrightarrow 2NH_3(g) + Heat$

How do the following changes influence the amount of the product ?

(a) Temperature decreases

(b) Pressure increases

(c) Ammonia produced is removed continuously from the system.

11. 2022 **2 MARKS**

(a) Which are the two compounds formed when Ammonium Chloride (NH_4Cl) is strongly heated ?

(b) Write the chemical equation for this reaction.

12. 2022 **1 MARK**

What happens to the rates of forward and backward reaction at equilibrium point ?

13. 2021 **4 MARKS**

The chemical equation of the industrial preparation of ammonia is given below.

$N_2(g) + 3H_2(g) \leftrightarrow 2NH_3(g) + Heat$

How does the following factors influence the rate of forward reaction ?

(a) Ammonia is removed from the system.

(b) Temperature is increased.

(c) Pressure is increased.

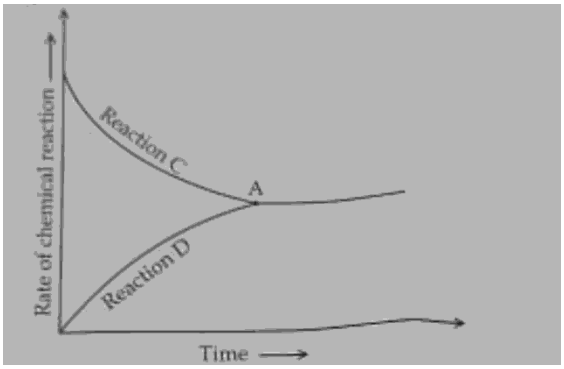
(d) More nitrogen is added.

14. 2021 **3 MARKS**

The graph for the reversible reaction $N_2(g) + 3H_2(g) \leftrightarrow 2NH_3(g)$ + Heat is given below.

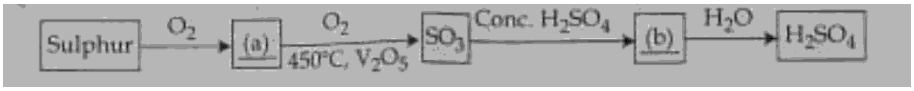
(a) Identify and write the chemical equation associated with 'Reaction C' and 'Reaction D'.

(b) What is the peculiarity of the point A ?



15. 2021 **2 MARKS**

The flow chart of the industrial preparation of sulphuric acid is given. Complete it.



16. 2021 **1 MARK**

Which is the drying agent used to remove moisture present in ammonia gas ?